COMPUTER NETWORKS LAB

Practical & Lab Examination Guide

CSE3131 • Cisco Packet Tracer • Theory • Step-by-step procedures • Commands • Verification

How to write each experiment in the lab exam

• Aim – what you are going to perform.

• Theory – explain the concept and its purpose.

• Requirements – router, switch, PCs, cables and Packet Tracer.

• Procedure – write the steps in order.

• Configuration – write commands or IP addresses.

• Output – write what you observed.

• Result – state whether the experiment was successful.

Prepared from the Computer Networks Lab (CSE3131) guide discussed in this chat. Use the interface names available on your selected Packet Tracer router model.

Experiment 1 – Introduction to Packet Tracer and Networking Devices

Aim

Aim: To study Cisco Packet Tracer and understand networking device components.

Theory (write in the lab exam)

Cisco Packet Tracer is network simulation software developed by Cisco. It allows users to create, configure, simulate and troubleshoot computer networks without physical hardware.

Important networking devices

Procedure

1. Open Cisco Packet Tracer.
2. Select Network Devices at the bottom.
3. Open Routers and place one router in the workspace.
4. Open Switches and place one switch.
5. Open End Devices and place two PCs.
6. Select Connections (lightning icon).
7. Choose Copper Straight-Through.
8. Connect PC0 to Switch0 and PC1 to Switch0.
9. Connect Switch0 to Router0.
10. Wait for the link indicators to become green.

Result

Studied Packet Tracer and identified networking devices and their functions.

Device	Function
Router	Connects different networks (Layer 3).
Switch	Connects devices in a LAN (Layer 2).
Hub	Broadcasts received data to all ports.
PC	Sends and receives data.
Server	Provides network services.
Access Point	Provides wireless connectivity.

Types of cables

• Straight-through: connects different device types, such as PC to switch.

• Crossover: traditionally connects similar devices, such as switch to switch.

• Serial: connects routers over serial links.

• Fiber optic: supports high-speed communication.

Experiment 2 – Network Utility Commands

Aim

Aim: To understand network utility commands used for monitoring and troubleshooting networks.

Theory (write in the lab exam)

Network utility commands help administrators check connectivity, IP configuration, active connections, MAC addresses, DNS information and packet routes.

Procedure

1. Start Windows and press Windows + R.
2. Type cmd and press Enter.
3. Execute the commands listed below one by one.
4. Observe the output of each command.
5. For Linux, use the Linux-specific commands listed below.

Result

Successfully studied network utility commands used for network monitoring and troubleshooting.

Commands and purpose

Command	Purpose
PING	Checks connectivity.
NETSTAT	Displays active connections and ports.
IPCONFIG	Displays Windows IP configuration.
IFCONFIG	Displays Linux interface configuration.
ARP	Displays IP-to-MAC mapping.
TRACERT	Shows the route taken by packets.
NSLOOKUP	Finds IP addresses using DNS.
PATHPING	Checks route, latency and packet loss.

```
ping google.com
ipconfig
ipconfig /all
netstat -a
netstat -r
netstat -e
arp -a
tracert google.com
nslookup google.com
pathping google.com
```

Linux commands:

```
ifconfig
traceroute google.com
```

Observations: PING shows reply time and TTL; IPCONFIG shows IPv4 address, subnet mask and gateway; ARP shows IP and MAC addresses; TRACERT shows intermediate routers; NSLOOKUP shows DNS resolution.

Result

Successfully studied network utility commands used for network monitoring and troubleshooting.

Experiment 3 – Star Topology Using Hub and Switch

Aim

Aim: To create a star topology using a hub and switch, configure IP addresses and test communication.

Theory (write in the lab exam)

Star topology is a network topology in which all computers are connected to a central device such as a hub or switch. All communication between devices passes through the central device. A switch forwards frames using MAC addresses, whereas a hub broadcasts data to all connected ports.

Procedure

1. Open Packet Tracer.
2. Select Switches and drag one 2960 switch into the workspace.
3. Select End Devices and drag six PCs into the workspace.
4. Select Connections (lightning icon).
5. Choose Copper Straight-Through.
6. Connect each PC's FastEthernet0 port to a different switch port.
7. Wait until the link indicators turn green.
8. Click PC0, then Desktop → IP Configuration.
9. Enter the IP addresses and subnet masks from the table below.
10. Repeat the IP configuration for all six PCs.
11. Open PC0 → Desktop → Command Prompt.
12. Run ping 192.168.1.2 and check for successful replies.
13. Repeat the ping test with another PC.
14. Optionally configure the switch hostname and password using the commands below.

Result

PC	IP address	Subnet mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0
PC4	192.168.1.5	255.255.255.0
PC5	192.168.1.6	255.255.255.0

Optional switch configuration

```
enable
configure terminal
hostname MySwitch
enable password cisco
exit
copy running-config startup-config
```

Verification commands

```
show running-config
show mac address-table
ping 192.168.1.2
```

Result

Successfully created star topology and verified communication using PING.

Experiment 4 – DHCP Configuration Using Router and Switch

Aim

Aim: To configure DHCP on a router to automatically assign IP addresses to PCs.

Theory (write in the lab exam)

DHCP stands for Dynamic Host Configuration Protocol. It automatically assigns IP addresses, subnet masks, default gateways and DNS information to devices. DHCP follows DORA: Discover (client searches), Offer (server offers an address), Request (client requests it), and Acknowledgment (server confirms it).

Procedure

1. Open Packet Tracer.
2. Place one 2911 router, one 2960 switch and two PCs.
3. Choose Copper Straight-Through.
4. Connect Router GigabitEthernet0/0 to Switch FastEthernet0/1.
5. Connect PC0 to Switch FastEthernet0/2.
6. Connect PC1 to Switch FastEthernet0/3.
7. Click Router → CLI and press Enter.
8. Configure the router interface using the commands below.
9. Configure the DHCP excluded-address and pool.
10. Click PC0 → Desktop → IP Configuration → DHCP.
11. Wait for an IP address to appear and repeat for PC1.
12. Verify the assigned addresses on the router.

Result

Router configuration

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp pool MY_POOL
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 8.8.8.8
exit
```

Verification

```
show ip dhcp binding
show ip dhcp pool
show ip interface brief
```

Expected example: PC0 may receive 192.168.1.11 and PC1 may receive 192.168.1.12. Gateway: 192.168.1.1. Subnet mask: 255.255.255.0. Actual leases can vary.

Result

Successfully configured DHCP and automatically assigned IP addresses to PCs.

Experiment 5 – Router Modes and Basic Switch/Router Commands

Aim

Aim: To understand router operating modes and execute basic router and switch commands.

Theory (write in the lab exam)

Cisco routers operate in different modes. User EXEC provides basic access; Privileged EXEC provides advanced commands; Global Configuration is used for overall configuration; Interface Configuration is used to configure an individual interface.

Procedure

1. Open Packet Tracer.
2. Place one router and one switch.
3. Connect them using a straight-through cable.
4. Click Router → CLI and press Enter.
5. Enter enable to move to privileged EXEC mode.
6. Enter configure terminal to move to global configuration mode.
7. Change the hostname.
8. Select GigabitEthernet0/0 and configure its IP address.
9. Enable the interface using no shutdown.
10. Set an enable secret password.
11. Exit configuration mode and inspect the configuration.
12. Save the configuration.

Result

Mode	Prompt	Purpose
User EXEC	Router>	Basic commands
Privileged EXEC	Router#	Advanced commands
Global Configuration	Router(config)#	Overall configuration
Interface Configuration	Router(config-if)#	Interface configuration

```
enable
configure terminal
hostname MyRouter
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
enable secret admin123
end
show running-config
show ip interface brief
copy running-config startup-config
```

Press Enter if asked for the destination filename.

Result

Successfully explored router modes and configured basic router and switch commands.

Experiment 6 – Static Routing Configuration

Aim

Aim: To configure static routing between two routers.

Theory (write in the lab exam)

Static routing is a routing technique in which the network administrator manually enters routes into the router's routing table. It does not automatically change routes when network topology changes. It is simple and predictable but requires manual updates.

Procedure

1. Open Packet Tracer and place two routers (R1 and R2).
2. Place two switches and two PCs.
3. Connect PC0 → Switch0 → R1 G0/0.
4. Connect R1 G0/1 to R2 G0/1 using Copper Cross-Over or Automatic connection.
5. Connect R2 G0/0 → Switch1 → PC1.
6. Configure PC0 as 192.168.1.2/24, gateway 192.168.1.1.
7. Configure PC1 as 192.168.2.2/24, gateway 192.168.2.1.
8. Configure R1 interfaces and its static route.
9. Configure R2 interfaces and its return static route.
10. Verify interfaces and routing tables.
11. From PC0, ping 192.168.2.2.

Result

Device	Interface	IP address
PC0	FastEthernet0	192.168.1.2
R1	G0/0	192.168.1.1
R1	G0/1	10.0.0.1
R2	G0/1	10.0.0.2
R2	G0/0	192.168.2.1
PC1	FastEthernet0	192.168.2.2

LAN subnet masks: 255.255.255.0. Router-to-router subnet mask: 255.255.255.252.

Configure R1

```
enable
configure terminal
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface g0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
```

```
exit
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

Configure R2

```
enable
configure terminal
interface g0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
interface g0/1
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

Verification

```
show ip route
show ip interface brief
ping 192.168.2.2
```

Remember: both routers need static routes. Configuring only one side is not enough.

Result

Successfully configured static routing between two different networks.

Experiment 7 – RIPv1 and RIPv2 Configuration

Aim

Aim: To configure RIP dynamic routing between multiple routers.

Theory (write in the lab exam)

RIP stands for Routing Information Protocol. It is a distance-vector dynamic routing protocol that selects routes using hop count. The maximum usable hop count is 15; 16 means unreachable. RIPv1 is classful and does not send subnet masks. RIPv2 is classless, supports subnet masks, VLSM and CIDR, and uses multicast updates.

Procedure

1. Build the same two-router, two-LAN topology used in Experiment 6.
2. Configure IP addresses on PCs and router interfaces exactly as in Experiment 6.
3. Enable all router interfaces using no shutdown.
4. On R1, enter the RIPv1 commands below.
5. On R2, enter the RIPv1 commands below.
6. To use RIPv2 instead, configure version 2 and no auto-summary on both routers using the alternative commands.
7. Verify learned routes and test a ping between PCs.

Result

RIPv1	RIPv2
Classful protocol	Classless protocol
Does not send subnet masks	Sends subnet masks
No VLSM support	Supports VLSM
Broadcast updates	Multicast updates
No authentication support	Supports authentication

RIPv1 – R1

```
enable
configure terminal
router rip
version 1
network 192.168.1.0
network 10.0.0.0
exit
```

RIPv1 – R2

```
enable
configure terminal
router rip
version 1
network 192.168.2.0
network 10.0.0.0
exit
```

RIPv2 – R1

```
router rip
version 2
no auto-summary
network 192.168.1.0
network 10.0.0.0
```

RIPv2 – R2

```
router rip
version 2
no auto-summary
network 192.168.2.0
network 10.0.0.0
```

Verification

```
show ip route
show ip protocols
ping 192.168.2.2
```

Routes learned through RIP are marked with R in the routing table. Configure RIP on both routers.

Result

Successfully configured RIPv1 and RIPv2 for dynamic routing.

Experiment 8 – OSPF Configuration and Troubleshooting

Aim

Aim: To configure OSPF routing between multiple routers.

Theory (write in the lab exam)

OSPF stands for Open Shortest Path First. It is a link-state dynamic routing protocol that selects the best route based on cost. OSPF uses Dijkstra's Shortest Path First algorithm. It supports hierarchical routing using areas; Area 0 is the backbone area.

Procedure

1. Build the same two-router, two-LAN topology used in Experiment 6.

2. Assign IP addresses to all PCs and router interfaces.

3. Configure subnet masks and default gateways.

4. Enable router interfaces using no shutdown.

5. On R1, enable OSPF process 1, set a router ID and advertise its LAN and transit networks in Area 0.

6. On R2, enable OSPF process 1, set a different router ID and advertise its LAN and transit networks in Area 0.

7. Verify OSPF neighbors, routing table and protocol information.

8. From PC0, ping 192.168.2.2.

Result

Configure R1

```
enable
configure terminal
router ospf 1
router-id 1.1.1.1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
exit
```

Configure R2

```
enable
configure terminal
router ospf 1
router-id 2.2.2.2
network 192.168.2.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
exit
```

Verification

```
show ip ospf neighbor
show ip route
show ip protocols
ping 192.168.2.2
```

Problem	Solution
Interface down	Use no shutdown.
Neighbor not forming	Check IP addresses and subnet masks.
Routes missing	Check OSPF network commands.
Wrong area	Configure matching area.
Ping failing	Check gateways and routing tables.

Routes learned through OSPF are marked with O. The OSPF process ID is locally significant; both routers do not have to use the same process ID.

Result

Successfully configured OSPF and established communication between different networks.

Experiment 9 – VLAN Configuration and Troubleshooting

Aim

Aim: To configure VLANs on a switch and assign ports to different VLANs.

Theory (write in the lab exam)

VLAN stands for Virtual Local Area Network. It logically divides a physical network into separate broadcast domains. Devices in the same VLAN can communicate at Layer 2. Devices in different VLANs require Layer 3 routing for communication.

Procedure

1. Open Packet Tracer and place one 2960 switch.

2. Place four PCs.

3. Connect all four PCs to the switch using Copper Straight-Through.

4. Configure PC0 as 192.168.10.2/24 and PC1 as 192.168.10.3/24.

5. Configure PC2 as 192.168.20.2/24 and PC3 as 192.168.20.3/24.

6. Click Switch → CLI and enter enable, then configure terminal.

7. Create VLAN 10 named HR.

8. Create VLAN 20 named SALES.

9. Assign Fa0/1 and Fa0/2 to VLAN 10 as access ports.

10. Assign Fa0/3 and Fa0/4 to VLAN 20 as access ports.

11. Verify using show vlan brief.

12. Ping PC1 from PC0; this should work.

13. Ping PC2 from PC0; this should fail without inter-VLAN routing.

14. If required, configure a trunk port on Fa0/5.

Result

PC	IP address	VLAN
PC0	192.168.10.2	10
PC1	192.168.10.3	10
PC2	192.168.20.2	20
PC3	192.168.20.3	20

Create VLANs

```
enable
configure terminal
vlan 10
name HR
exit
vlan 20
name SALES
exit
```

Assign ports to VLAN 10

```
interface range fa0/1-2
switchport mode access
switchport access vlan 10
exit
```

Assign ports to VLAN 20

```
interface range fa0/3-4
switchport mode access
switchport access vlan 20
exit
```

Optional trunk configuration

```
interface fa0/5
switchport mode trunk
```

Verification

```
show vlan brief
```

Troubleshooting: check port assignment with show vlan brief, IP addresses and subnet masks, trunk configuration, and whether devices are connected to the correct ports.

Result

Successfully created VLAN 10 and VLAN 20 and verified network segmentation.

Experiment 10 – NAT Protocol Configuration and Troubleshooting

Aim

Aim: To configure Network Address Translation and verify private-to-public IP translation.

Theory (write in the lab exam)

NAT stands for Network Address Translation. It translates private IP addresses into public IP addresses and vice versa. It helps conserve public IP addresses and hides internal IP addressing from external networks. Static NAT maps one private IP to one fixed public IP; Dynamic NAT uses a public pool; PAT allows multiple private IPs to share one public IP using port numbers.

Procedure

1. Open Packet Tracer.
2. Place one router, two switches, two PCs and one server.
3. Connect PC0 and PC1 to Switch0; connect Switch0 to Router G0/0.
4. Connect Router G0/1 to Switch1 and connect the server to Switch1.
5. Configure PC0 as 192.168.1.2/24 and PC1 as 192.168.1.3/24.
6. Configure Router G0/0 as 192.168.1.1/24 and Router G0/1 as 203.0.113.1/24.
7. Configure Server as 203.0.113.2/24 with gateway 203.0.113.1.
8. Mark G0/0 as NAT inside and G0/1 as NAT outside.
9. Create an access list permitting the internal network.
10. Configure PAT overload using the outside interface.
11. From PC0, ping 203.0.113.2.
12. Verify NAT translations and statistics on the router.

Result

Device	IP address
PC0	192.168.1.2
PC1	192.168.1.3
Router G0/0	192.168.1.1
Router G0/1	203.0.113.1
Server	203.0.113.2

Subnet mask: 255.255.255.0. PC default gateway: 192.168.1.1. Server gateway: 203.0.113.1.

Configure inside interface

```
enable
configure terminal
interface g0/0
ip address 192.168.1.1 255.255.255.0
ip nat inside
no shutdown
exit
```

Configure outside interface

```
interface g0/1
ip address 203.0.113.1 255.255.255.0
ip nat outside
no shutdown
exit
```

Create access list and configure PAT

```
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface g0/1 overload
```

Verification

```
show ip nat translations
show ip nat statistics
ping 203.0.113.2
```

Troubleshooting: check inside/outside interfaces, access-list configuration, router interface IP addresses, gateway configuration and NAT translations.

Result

Successfully configured PAT and verified private-to-public IP address translation.

Experiment 11 – Nmap, Wireshark and Network Scanner

Aim

Aim: To discover network devices, identify open ports and analyze network traffic.

Theory (write in the lab exam)

Nmap (Network Mapper) is an open-source network scanning tool used to discover hosts, open ports, services and operating systems. Wireshark is a network protocol analyzer used to capture and examine packets in real time. Network scanners discover devices and identify IP and MAC addresses.

Procedure

1. For Nmap, specify an authorized target IP address or range.
2. Run a host discovery or port scan.
3. Review active hosts, port states and service information.
4. For Wireshark, open the application and select the active network interface.
5. Start packet capture.
6. Generate traffic by opening a website or running ping.
7. Apply display filters such as icmp, tcp, dns or http.
8. Inspect packet details, stop capture and save the file in PCAP format.
9. For a network scanner, specify the IP range and start scanning.
10. Record discovered devices and their IP/MAC addresses.

Result

Nmap commands – only scan systems you own or are authorized to test

```
nmap 192.168.1.1
nmap -sV 192.168.1.1
nmap -sn 192.168.1.0/24
```

Wireshark display filters

```
icmp
tcp
dns
http
```

Result

Successfully studied network scanning and packet analysis tools.

Experiment 12 – Network Security, Threats and Vulnerabilities

Aim

Aim: To understand network security threats, attacks and methods of protection.

Theory (write in the lab exam)

Network security protects data and devices from unauthorized access, misuse and attacks. A threat is a potential danger to a network, whereas a vulnerability is a weakness that can be exploited.

Procedure

1. Identify network devices and important data that need protection.
2. Identify possible vulnerabilities.
3. Study potential network attacks.
4. Evaluate the impact of attacks.
5. Apply suitable security measures.
6. Monitor network activity.
7. Prepare a security assessment report.

Result

Attack	Explanation
DoS	Makes a service unavailable by overwhelming it.
MitM	Intercepts communication between two parties.
Phishing	Tricks users into revealing sensitive information.
Malware	Malicious software that damages or steals data.

Security mitigation methods

• Firewalls.

• IDS/IPS.

• Antivirus software.

• Regular software updates.

• Network monitoring.

• User awareness training.

• Strong passwords and encryption.

• Secure device configurations.

Result

Studied network threats, attacks, vulnerabilities and security mitigation techniques.

Experiment 13 – Case Study – Building and Scaling a Network

Aim

Aim: To design a small network, expand it into a larger network and troubleshoot connectivity problems.

Theory (write in the lab exam)

Network scaling means increasing the size and capacity of a network to support additional users, departments and devices while maintaining reliable communication.

Procedure

1. Phase 1 – Small network: place one router, one switch and two PCs.

2. Connect the PCs to the switch and the switch to the router.

3. Configure DHCP and test communication using ping.

4. Phase 2 – Scale the network by adding two departments, a file server, a backup router, additional PCs, printers and network devices.

5. Configure separate subnets for departments.

6. Configure VLANs for traffic segmentation.

7. Implement OSPF for communication between subnets.

8. Add a file server for centralized storage.

9. Add a backup router for redundancy.

10. Phase 3 – Simulate and troubleshoot connectivity, IP conflict and performance problems.

Result

Problem	Solution
PC connectivity failure	Correct VLAN assignment.
IP conflict	Assign unique IP addresses and adjust DHCP scope.
Slow network	Review traffic and configure QoS where appropriate.
Routing problem	Check routing table.
Interface down	Use no shutdown.

Result

Successfully studied network design, expansion and troubleshooting.

Quick Revision – Important Packet Tracer Commands

Purpose	Command
Enter privileged mode	enable
Configuration mode	configure terminal
Set hostname	hostname R1
Configure interface	interface g0/0
Assign IP	ip address IP MASK
Enable interface	no shutdown
Check interfaces	show ip interface brief
Check routing	show ip route
Save configuration	copy running-config startup-config
Create VLAN	vlan 10
Check VLAN	show vlan brief
Configure RIP	router rip
Configure OSPF	router ospf 1
Configure static route	ip route NETWORK MASK NEXT-HOP
Check DHCP	show ip dhcp binding
Check NAT	show ip nat translations
Check OSPF neighbors	show ip ospf neighbor

Common Mistakes That Cause Packet Tracer Experiments to Fail

• Forgetting no shutdown on router interfaces.

• Giving incorrect subnet masks.

• Setting the wrong default gateway on PCs.

• Connecting the wrong router interfaces.

• Configuring routing on only one router.

• Forgetting to advertise a network in RIP or OSPF.

• Assigning incorrect switch ports to VLANs.

• Forgetting to configure NAT inside and outside.

• Testing before interfaces turn green.

• Forgetting to save configurations.

Practical Exam Preparation – Priority

Priority	Experiments
Very high	Static Routing, RIP v1/v2, OSPF, VLAN, NAT
High	DHCP, Star Topology, Router Commands
Medium	Network Utilities, Packet Tracer Introduction, Nmap/Wireshark, Network Security, Case Study

Important note: The lab manual's detailed program headings identify Program 3 as Star Topology, Program 4 as DHCP and Program 5 as Router Modes. Follow the detailed program headings when writing your practical. Interface names can differ by router model. If your router shows GigabitEthernet0/0/0 instead of GigabitEthernet0/0, use the interface name shown in Packet Tracer.